

Live Demonstration: A Pulmonary Conditions Monitor Based on Electrical Impedance Tomography measurement

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Abstract— In this demonstration, we present a non-invasive, real-time lung imaging system based on the electrical impedance tomography (EIT) technique. EIT is a medical imaging technique based on the electrical properties, i.e. resistivity and permittivity of tissues and organs. This demo is based on an EIT system-on-chip that utilizes frequency division multiplexing scheme to improve the throughput by 10, allowing clinicians to identify and prevent mechanical pulmonary injury during lung ventilation.

I. INTRODUCTION

Acute lung injury is a very common complication in the intensive care unit. Mechanical pulmonary injury can be prevented by lung protective ventilation. Although computed tomography (CT) provides detailed regional information about the lung, it is not suitable for continuous regional lung monitoring and cannot be used to guide the routine adjustment of ventilator settings [1, 2].

Compared with CT, electrical impedance tomography (EIT) can achieve non-invasive, radiation-free monitoring of regional lung function. The EIT measurement is done by applying 16 electrodes around the chest. Among 16 electrodes, one pair of electrode is used to inject an alternative current to the body while the rest electrodes form 13 pairs of reading channel to record voltages generated by the injected current. The current injection and voltage recording process rotate from the first pair electrodes to the last pair of electrode, generating 16×13 voltage readings. EIDORS software is used to construct the EIT image with 64×64 pixels.

II. DEMONSTRATION SETUP

The demonstration is based on a recently developed EIT SoC chip [3] . and a real-time data extraction and image reconstruction platform. As shown in Fig. 1, we use a customized circular tank, filled with water and nylon blocks to model the thorax. The prototype EIT measurement system consists of the EIT SoC for bioimpedance acquisition, a data acquisition (DAQ) card to record data stream from the EIT chip, a graphical user interface (GUI) to visualize the lung image and to configure the EIT chip in real-time based on the application scenario. A Matlab program converts the data from EIT chip readout into voltage values. The setup successfully demonstrates the low-power performance of B.Lius chip as well as the advantages of the embedded algorithm [2, 3] .

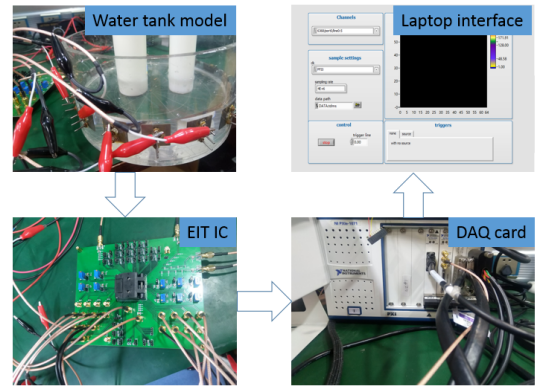


Fig. 1: Experimental setup of our EIT measurement system

III. VISITOR EXPERIENCE

During the demonstration, the visitors can shift the plastic object within the water-tank and observe the corresponding changes in the bitmap image in real time. Through interaction with our GUI, visitors could change important parameters like sampling rate and the number of channels, which determines the resolution and refreshing rate of the bitmap imaging.

IV. ACKNOWLEDGMENT

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REFERENCES

- [1] H. Li, B. Liu, Y. Li, G. Wang, and Y. Lian, “A High Accuracy and High Sensitivity System Architecture for Electrical Impedance Tomography System,” in *Proc. IEEE Asia Pacific Conf. on Circuits, and Syst. (APCCAS)*, October 2018, pp. 1–4.
- [2] B. Liu, C.H. Heng, G. Wang, and Y. Lian, “On-chip Data Compression Scheme for Lung EIT Signal Acquisition and Recovery,” in *Proc. IEEE Int. Conf. on Digital Signal Process. (DSP)*, November 2018, pp. 1–4.
- [3] B. Liu, G. Wang, Y. Li, H. Li, Y. Gao, L. Zeng, Y. Ma, Y. Lian, and C.H. Heng, “A 13-Channel 1.53mW 11.28mm² Electrical-Impedance-Tomography SoC Based on Frequency Division Multiplexing with 10x Throughput Reduction,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, February 2019, pp. 1–2.